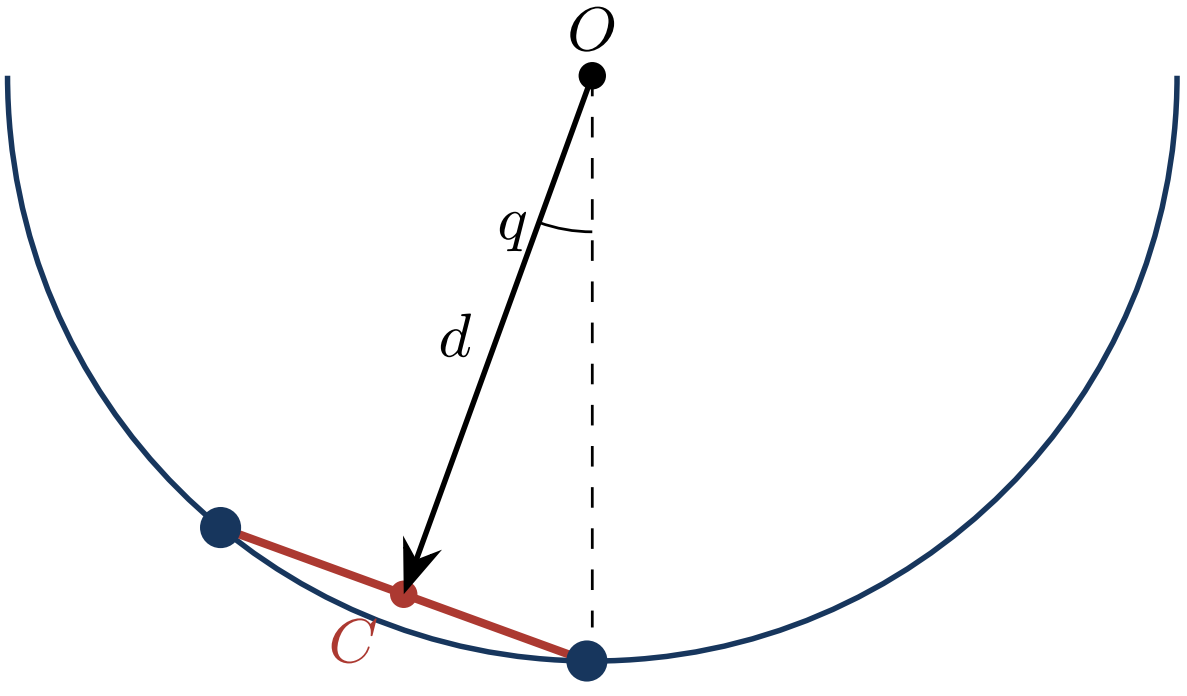

$$K = mR^2\dot{q}^2$$

$$U = 2mgd(1 - \cos q)$$